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(54) Title: A HAPTIC GLOVE

(57) Abstract: The present invention relates to haptic gloves that provide the user with a sense of touch for use in virtual reality applications. It comprises features for transferring the touch function to the user in the most real way possible in the virtual reality environment.

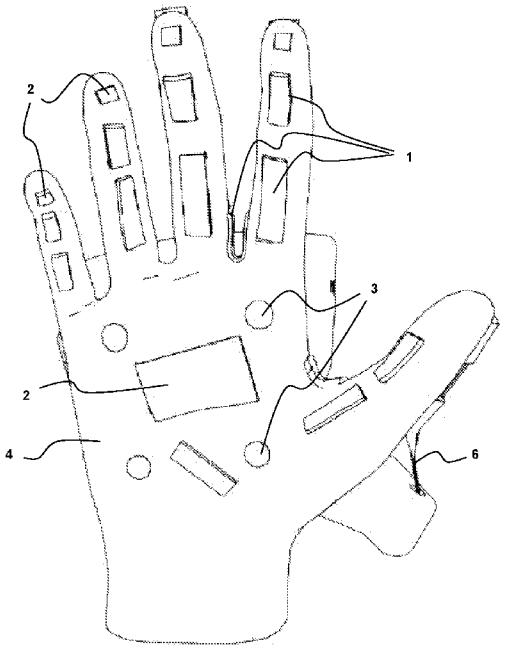


Figure – 1

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A HAPTIC GLOVE

Technological Field:

The present invention relates to haptic gloves that provide the user with a sense of touch and hold for use in virtual reality applications.

State of the Art:

The digital version of an imaginary or real environment is presented to users in virtual reality applications. Equipment such as VR glasses and mobile cabins are used so as to achieve this. The invention, which is the subject of the description, describes the haptic glove that aims to enable users to perform the touch function in the virtual reality environment.

There are haptic glove designs in the known art. The disadvantage of these glove designs is that they do not fully pass the feeling of touching and holding to the user. This is because the haptic sensation depends on so many parameters. Flexibility, temperature, movement, vibration, contact points are among the parameters that affect touching and holding. In the known art, it is not possible to reach the real feeling of touch, since this feeling is tried to be provided only by vibration.

A haptic glove design is described in the European Patent document numbered US2022179491 encountered in the literature search. The abovementioned glove does not have a motor, rope system and heating system that creates tension in the fingers. It tries to make the haptic sensation felt by using only flexibility and vibration parameters. As a result, the haptic sensation cannot be conveyed to the user at a sufficient level of reality.

Consequently, a new haptic glove is required in which the state of the art is exceeded, the disadvantages are eliminated.

Brief Description of the Invention:

The invention is a haptic glove which exceeds the state of the art, eliminates the disadvantages and has some additional features.

The aim of the invention relates to a haptic glove to convey the touch function to the user in the most real way possible in the virtual reality environment.

Another aim of the invention is to present a haptic glove that improves the touch and holding experience in the virtual reality environment by simultaneously transferring the temperature, vibration, flexibility, tension and movement parameters to the user.

Description of the Figures:

The present invention will be described with reference to the accompanying drawings, thus the characteristics of the invention will be understood clearly. However, the aim of this is not to limit the invention with such certain embodiments. On the contrary, it is aimed to cover all alternatives, amendments and equivalents which may be contained in the field defined by the accompanying claims. It is to be understood that the details shown are only shown for the sake of illustrating the preferred embodiments of the present invention and presented for both illustrating the methods and for providing description of the rules of the invention and the conceptual features of the invention to be easily understood. In these figures;

Figure – 1 is the palm view of the inventive haptic glove.

Figure – 2 is the external view of the inventive haptic glove.

The figures which enable to clarify this invention are enumerated as mentioned in the attached figure and they are given with their names herein below.

Description of the References:

1. Flexibility measurement sensor
2. Temperature pads
3. Vibration motors
4. Glove textile
5. Vibrating fingertip part
6. Pulling rope
7. Pulling rope guides
8. Electronic system box
9. Microcontroller based mainboard
10. Tension motor
11. Motor motion tracking system

Description of the Invention:

In this detailed description, the inventive haptic glove is described by means of examples only for clarifying the subject matter such that no limiting effect is created.

In Figure 1, the view showing the palm part of the haptic glove, which is the subject of the invention, is given. Accordingly, the haptic glove comprises temperature pads (2) so as to create a feeling of heat in a safe range for human skin in the middle of the fingertips and palm. There are vibration motors (3) in the palm so as to create a physical touch or warning sensation. Flexibility measurement sensors (1) are used in the glove so as to detect finger movements, to measure the bending and movement rates of the finger joints or between the fingers or the connected surfaces. Therefore, it is positioned on the glove such that it coincides with the knuckles or interdigital areas. The glove textile (4) used

in the invention is made of technical fabrics and electronic textile structures and a material selected according to hand ergonomics and comfortable movement.

In Figure 2, the outer surface of the haptic glove, which is the subject of the invention, is shown. Accordingly, there are vibrating fingertip parts (5) on the outer fingertips of the glove. Vibrating fingertips (5) may be provided at least one for each finger. The vibrating fingertip parts (5) perform the tasks of both physical touching and warning sensations on the fingertips and feeling the effect of the pulling ropes (6) on the fingertips. The pulling ropes (6) mentioned here are lines whose tension can be adjusted such that the solid state of the objects held in the virtual reality environment affects finger movements and haptic sensation. The position of the pulling rope (6) on the finger forms in the glove is provided by the pulling rope guides (7). In the glove subject to the invention, there can be a large number of pulling rope guides (7), at least one of which guides the pulling ropes (6). In the present invention, the pulling ropes (6) are stretched and oscillated by the motor motion tracking systems (11) driven by the tension motors (10). The tension motors (10) are the drive equipment that adjusts the tension of the pulling ropes (6) such that the interactions with the virtual objects in the environment are physically created on the fingers. The motor motion tracking system (11) monitors the movements of the tension motors and ensures that they are used in tension and finger movements. In the present invention, the number of tension motor (10) and motor motion tracking system (11) is equal to the number of fingers to be stretched. As seen in Figure 2, three fingers of the glove are stretched. There are three tension motors (10) and three motor motion tracking systems (11) for this purpose. However, it is also possible to stretch different fingers independently of each other with a single stretching motor (10) and a single motor motion tracking system (11) depending on preference. There is no limitation in this regard in the present invention.

In the present invention, the temperature, flexibility measurement, vibration and stretching movements on the glove are all controlled via the microcontroller-based mainboard (9) in an electronic system box (8). The electronic system box

(8) is the structure that enables the electronic and electromechanical systems to be safely positioned on the glove and the tracking device to be mounted on the glove, which ensures that the glove is tracked in the movement space. Microcontroller-based mainboard (9) describes the system that collects and processes data from sensors, is in two-way communication with computer software, and manages virtual reality stimuli.

The number and positions of temperature pads (2), flexibility measurement sensors (1), vibration motors (3), vibrating fingertips (5) used in the invention may vary. There is no limitation regarding these. Likewise, the number of pulling ropes (6) and pulling rope guides (7) can vary, and there must be at least one in each finger to which haptic sensation is desired.

Finally, the above-mentioned electronic line connections are made with electronic textiles on the glove side. Electronic textiles provide comfort, lightness and real usability to the final product.

CLAIMS

1. A haptic glove that provides the user with a sense of touch for use in virtual reality applications, **characterized in that**, it comprises the following;

- at least one glove textile (4) made of technical fabric material, imitating the finger and palm design of the user,
- at least one temperature pad (2) to ensure that said glove textile (4) creates a feeling of heat in a safe range for human skin,
- Pulling rope (6), the tension of which can be adjusted by at least one motor to affect the haptic sensation, at least one on each finger for which the haptic sensation is desired,
- flexibility measurement sensors (1) to detect finger movements, to measure the bending and movement rates of the finger joints or between the fingers or the connected surfaces,
- microcontroller-based mainboard (9) that collects and processes data from sensors, is in two-way communication with computer software, and manages hardware with virtual reality stimuli.

2. A haptic glove according to Claim 1, **characterized in that**, it comprises a large number of temperature pads (2) located on the palm and fingertips of the glove.

3. A haptic glove according to Claim 1, **characterized in that**, it comprises pulling rope guides (7) that provide the position of the pulling ropes (6) on the finger.

4. A haptic glove according to Claim 1, **characterized in that**, it comprises at least one electronic system box (8) that enables the electronic and electromechanical systems to be safely positioned on the glove and the tracking device to be mounted on the glove, which ensures that the glove is tracked in the movement space.

5. A haptic glove according to Claim 1, **characterized in that**, it comprises vibration motors (3) in the palm section so as to create a physical touch or warning sensation.

6. A haptic glove according to Claim 1, **characterized in that**, it comprises vibrating fingertips (5) that perform the tasks of both physical touching and warning sensations on the fingertips and feeling the effect of the pulling ropes (6) on the fingertips.

7. A haptic glove according to Claim 1, **characterized in that**, it comprises at least one tension motor (10) that adjusts the tension of the pulling ropes (6) such that the interactions with the virtual objects in the environment are physically created on the fingers.

8. A haptic glove according to Claim 1, **characterized in that**, it comprises at least one motor motion tracking system (11) that monitors the movements of the tension motors and ensures that they are used in tension and finger movements.

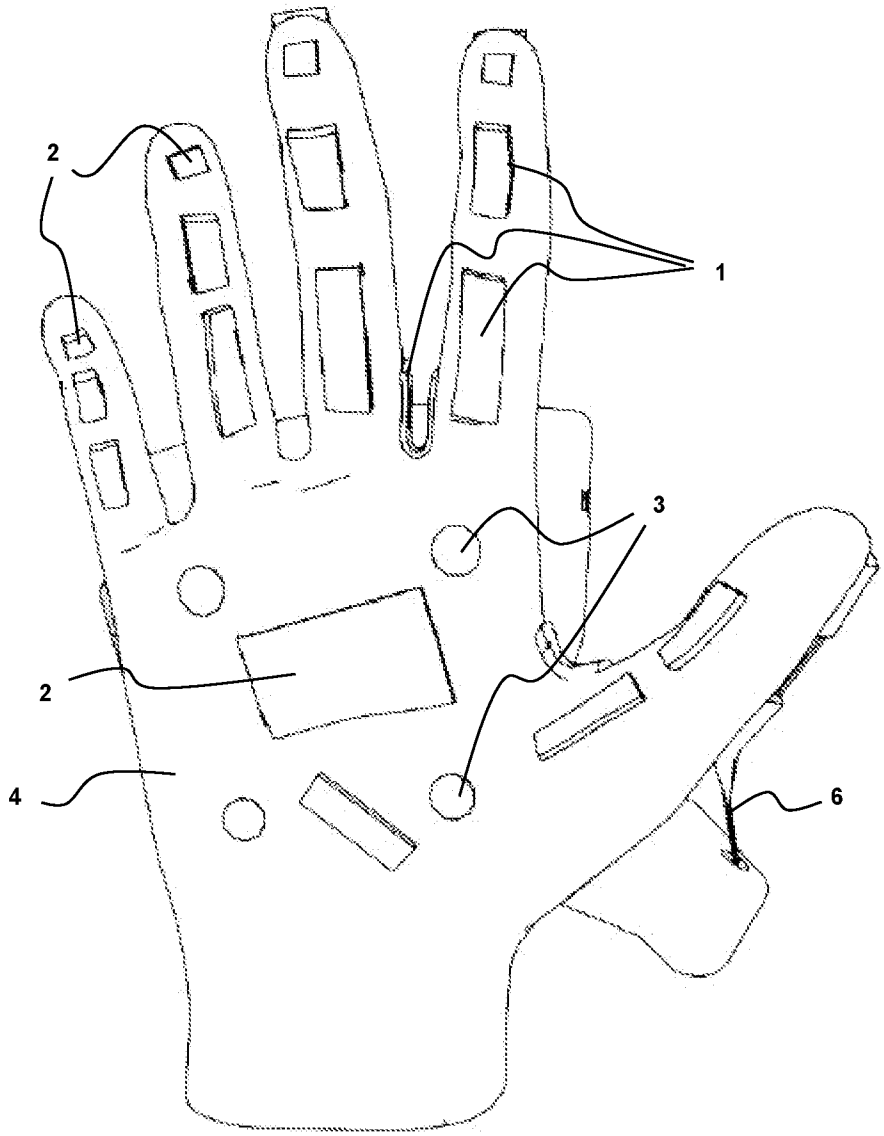
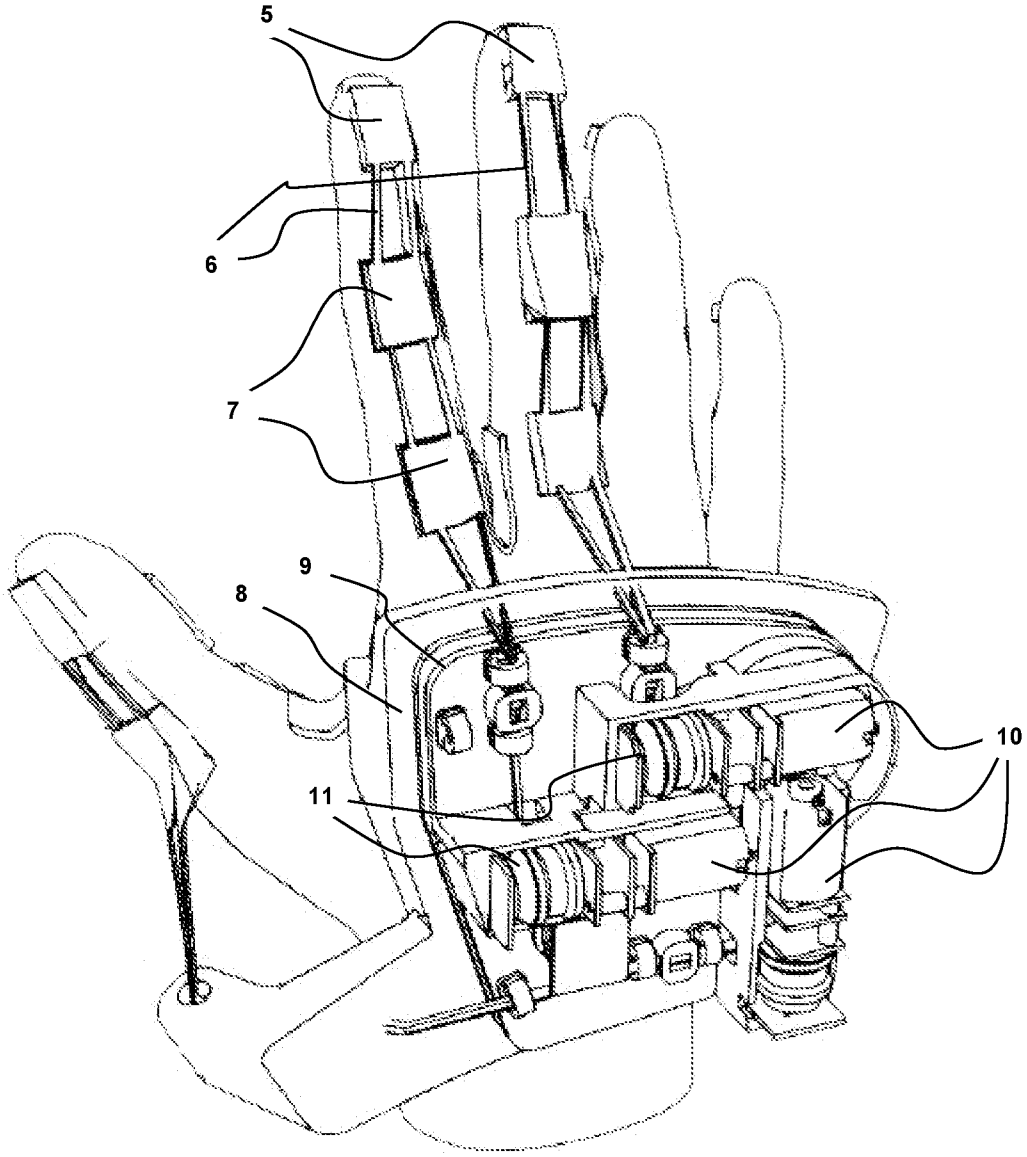


Figure – 1



Figure– 2

INTERNATIONAL SEARCH REPORT

International application No.
PCT/TR2023/050733

A. CLASSIFICATION OF SUBJECT MATTER

G06F 3/01 (2006.01)i

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

G06F 3/01

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPODOC

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2019101981 A1 (APPLE INC [US]) 04 April 2019 (2019-04-04) Abstract, Figures, Paragraphs [0005],[0020]-[0042],[0048]-[0079]	1-8
X	US 2021141452 A1 (HUMAN MODE L L C [US]) 13 May 2021 (2021-05-13) Whole document	1-8
X	US 2018284896 A1 (VRGLUV LLC [US]) 04 October 2018 (2018-10-04) Whole document	1-8

☐ Further documents are listed in the continuation of Box C. ☑ See patent family annex.

* Special categories of cited documents:
"A" document defining the general state of the art which is not considered to be of particular relevance
"D" document cited by the applicant in the international application
"E" earlier application or patent but published on or after the international filing date
"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)
"O" document referring to an oral disclosure, use, exhibition or other means
"P" document published prior to the international filing date but later than the priority date claimed
"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention
"X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone
"Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art
"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
26 March 2024	**26 March 2024**

Name and mailing address of the ISA/TR	Authorized officer
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INTERNATIONAL SEARCH REPORT
Information on patent family members

International application No.
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Patent document cited in search report	Publication date (day/month/year)	Patent family member(s)	Publication date (day/month/year)
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